

Forensic Investigation Report

Case ID	CASE_4CFDEC4197	Report ID	RPT-4CFDEC4197-4320E361
Generated (UTC)	2026-07-25T14:41:27.464Z	Report version	v2

Every statement in this report references stored forensic findings; no content is generated outside computed results. Section "Report Provenance & Integrity" lists the SHA-256 digests of the source artifacts.

Executive Summary

- Case CASE_4CFDEC4197 contains 1 evidence item(s), each acquired under SHA-256 chain-of-custody verification.
- This case is linked to 1 other case(s) through shared entities: CASE_4D7CD3FD4E [cross_case_correlation.json].
- The weighted correlation engine found 0 related evidence pair(s) out of 0 analysed [correlation_analysis.json].

Scope & Methodology

- **objective:** Acquire, verify, correlate and reconstruct the digital evidence for this case, and derive investigative leads (suspect anchors, campaigns, cross-case links) strictly from stored, hash-verified artifacts.
- **evidence scope:** 1 evidence item(s) acquired through the CIIS intake pipeline under SHA-256 chain-of-custody control.
- **methodology:**
 - Phase 1 - Acquisition & OCR: PaddleOCR PP-OCRv5 text extraction with per-item confidence scoring; SHA-256 fingerprint recorded at intake and re-verified at read.
 - Phase 1 - Forensics: metadata/EXIF consistency, forgery signals, logo detection and evidence-confidence scoring stored per item under storage/forensics/.
 - Phase 2 - Correlation: weighted entity-overlap engine scoring every evidence pair; results in correlation_analysis.json.
 - Phase 2 - Cross-case: shared-entity matching against every other analysed case (cross_case_correlation.json).
 - Phase 2 - Campaigns / Suspects / Timeline: clustering, anchor derivation and event reconstruction over the correlated evidence set.
 - Threat intelligence: URL/domain indicators scored by the configured provider (static indicator file, or the trained phishing classifier when CIIS_ML_THREAT_INTEL=1); per-indicator results in 'Model Prediction Results'.
 - Reporting: this document is assembled exclusively from the stored outputs above; it contains no free-text generation.
- **reproducibility:** Re-running the analysis against the same stored evidence reproduces every figure herein; artifacts are versioned and never overwritten.

Case Overview

- **case id:** CASE_4CFDEC4197
- **evidence count:** 1
- **first evidence:** 2026-07-25T14:41:19.994Z
- **last evidence:** 2026-07-25T14:41:19.994Z
- **file types:**
 - url

Evidence Summary

Evidence	File	Acquired	OCR conf.	Integrity
EVID_00006	http_shared-scam-domain.test_pay.url	2026-07-25T14:41:19	100%	VERIFIED

SHA-256 digests for each item are listed in the Appendix (chain of custody).

Correlation Analysis

- **pair count:** 0
- **related pair count:** 0
- **strength distribution:**
- **top relationships:**
 - none

Cross-Case Correlation

- **related case count:** 1
- **related case ids:**
 - CASE_4D7CD3FD4E
- **links:**
 - **other case id:** CASE_4D7CD3FD4E
 - **relationship strength:** STRONG
 - **match confidence:** 0.5699
 - **match reason:** Shares 2 entity(ies) with CASE_4D7CD3FD4E: domain shared-scam-domain.test, url http://shared-scam-domain.test/pay
- **matched entities:**
 - **entity type:** domains
 - **value:** shared-scam-domain.test
 - **this evidence ids:**
 - EVID_00006
 - **other evidence ids:**
 - EVID_00007
 - **entity type:** urls
 - **value:** http://shared-scam-domain.test/pay
 - **this evidence ids:**
 - EVID_00006
 - **other evidence ids:**
 - EVID_00007

Campaign Analysis

- **campaign count:** 0
- **unclustered evidence:**
 - EVID_00006
- **campaigns:**
 - none

Timeline Analysis

- **summary:** 1 event(s) spanning 0.0 hour(s); 0 timestamp(s) unresolved.
- **stage progression:**
 - none
- **progression consistent:** True
- **milestones:**
 - **timestamp:** 2026-07-25T14:41:19.994000+00:00
 - **description:** Investigation start - first reconstructed evidence event
- **critical events:**
 - none

Suspect Assessment

No suspect anchors were derived from the evidence.

Threat Intelligence Summary

- intel available: 1.0
- indicators checked: 2.0
- malicious indicators: 2.0
- evidence with threats: 1.0
- threat evidence ratio: 1.0

Model Prediction Results

Indicator	Verdict	Risk	Confidence	Source model
http://shared-scam-domain.test/pay	MALICIOUS	79	79%	ml:xgboost

http://shared-scam-domain.test/pay

domain: shared-scam-domain.test · SSL: UNKNOWN · SPF: missing · DMARC: missing · trust: 50/100

- The hybrid decision engine confirmed this URL is phishing based on agreement between the ML model and threat indicators.
- AI model classified this URL as phishing with 100% confidence.
- SSL status could not be verified.
- The URL does not use HTTPS, meaning data is transmitted without encryption.
- The domain is missing SPF and DMARC email-authentication record(s), which legitimate organisations typically configure.
- Domain label 'test' is within edit distance 2 of protected brand 'tweet'.

- indicators classified: 1
- flagged malicious: 1

Evidence Quality Summary

- mean image quality: 0.0
- mean evidence confidence: 0.0
- mean forgery score: 0.0
- max forgery score: 0.0
- mean ocr confidence: 1.0
- hash verified count: 1.0

Metadata Summary

- note: No Phase-1 metadata reports stored for this case.

Investigation Statistics

- entity statistics:
 - domains: 1
 - urls: 1
- campaign statistics:
 - campaign count: 0.0
 - largest campaign size: 0.0
 - clustered evidence: 0.0
 - unclustered evidence: 1.0
 - mean campaign confidence: 0.0
- timeline statistics:
 - event count: 1.0
 - resolved event count: 1.0
 - unresolved event count: 0.0
 - inferred event count: 1.0
 - stage count: 0.0
 - critical event count: 0.0
 - timeline span hours: 0.0

- **correlation statistics:**
 - **pair count:** 0.0
 - **related pair count:** 0.0
 - **mean confidence:** 0.0
 - **max confidence:** 0.0

Confidence Analysis

- **note:** No Phase-1 confidence scores stored for this case.

Investigation Conclusion

- 1/1 evidence item(s) passed SHA-256 chain-of-custody verification.

Recommendations

- Case priority: MEDIUM (30.8/100) - Standard queue: process in normal rotation while monitoring for new linked evidence.

Report Provenance & Integrity

- **report id:** RPT-4CFDEC4197-4320E361
- **generated at:** 2026-07-25T14:41:27.464Z
- **generator:** CIIS Phase-2 reporting module (template-over-data; no free-text generation)
- **evidence set digest:** 30154e0b4fa8b29d8178ef0a1ee9384b677bea00bd4af2b601145b7d9c29305a
- **evidence set digest note:** SHA-256 over the sorted SHA-256 digests of every evidence item; any change to the evidence set changes this value.

Source artifact digests (SHA-256):

correlation_analysis_v2.json: 50b29a44571d5925f184abdde30f2df7c5842bc6df7d5801bb6bdfe785a1c6df
cross_case_correlation_v2.json: 2193b42245aa545dd6bce21855dc83c9d764498dfffb8b955329f64e1aa6e462d
campaign_analysis.json: 2c705fa0165570207686e21707dbbb05836bb1da772caff43f7da15f10a10620
suspect_assessment.json: 5962fe553a6028fd0d6e48e154417ba6d599d0d5ae431da5e4980ca4f60c49da
timeline_analysis_v2.json: c4529bf78d6b644fddded5ac69bc8a3a8fd37cac89d380f62b34f556a8ea6b36
analytics.json: 76205ca606995dfd9bddb30c8ef91c9b23a0da3bbad04e28453a16a5b1bc4e5e
case_priority.json: e54606fce7c9bbb5854abc30155b8c372eee7f2b78f036e1fe8b07e7d54a10cb
graph_v2.json: 12cd248b4abb07c0a98a31e7a722ea3cfd3d7503d0d355ba8d5f743a37db2bb6

Appendix

- **chain of custody:**
 - **evidence id:** EVID_00006
 - **sha256:** 08dcf69e8c1888d919ce088901076b95b3badf8d840a871a461d622d679ac619
 - **upload time:** 2026-07-25T14:41:19.994Z
 - **status:** processed
- **stored artifacts root:** storage/investigation/<CASE_ID>/
- **phase1 artifacts root:** storage/forensics/<EVIDENCE_ID>/

Prepared by (system)

Cybercrime Investigation Intelligence System
Automated Phase-2 reporting module

Reviewed by (investigator)

Date:

Date:

Signature: